

Reading 2: Due Wednesday before class

Assignment 2: Due Sunday 10pm

Test 1: Wednesday after reading week

Recap

Definition:

An integer n is even iff $\exists k(n = 2k)$

Definition:

An integer is odd iff $\exists k(n = 2k + 1)$

Consequence $\forall x(n \text{ is even OR } n \text{ is odd})$

Use long division to get a remainder of 0 or 1 when dividing by 2

Also

$\forall n \neg(n \text{ is even AND } n \text{ is odd})$

(Wait until proof by contradiction)

Proof techniques include:

- Direct proof
- Proof by contraposition
- Proof of contradiction
- Proof by cases
- and more.

Direct Proof

We use known facts and rules of inference until we reach the desired conclusion.

Example

Proposition:

If n and m are even, then $m + n$ is even.

To prove an implication,

$$p \rightarrow q$$

we can start with the hypothesis p and work to get q .

In this case, there is an implicit universal quantification

$$\forall n \forall m$$

So we're going to need to end with a universal generalization,

We need to keep m and n arbitrary, (We can't use any facts about m and n except their definitions)

Our proof could go like this

1. m is even $\wedge$ n is even (hypothesis)
2. m is even (simplification of 1.)
3. $\exists k(m = 2k)$ (definition of even applied to 2.)
4. $m = 2l$ for some l (existential instantiation of 3.) (Note: we do not reuse the name k .)
5. n is even
6. $\exists k(n = 2k)$
7. $n = 2o$ for some o

(Exercise: annotate 5 through 7)

8. $m + n = 2l + 2o$ (substitution of equals)
 - $\equiv 2(l + o)$ (algebra)
9. $\exists k(m + n = 2k)$ (existential generalization of 8. using the examples $l + o$)
10. $m + n$ is even (definition of even)

Conclusion: Thus if m is even and n is even, then $m + n$ is even.

Check that m and n really were arbitrary, thus the implied quantification applies.

Anatomy of a proof

- Number every single statement
- Justify every statement
- State a conclusion

Proof by contraposition

Sometimes it is easier to prove $\neg q \rightarrow \neg p$ then to prove $p \rightarrow q$

Since these are equivalent do the implication that is easier.

Proposition

If m and n are integers and mn is even, then m is even or n is even. (This is a special case of Euclid's Lemma (see chapter 4))

Claim

We write the claim as an implication:

1.
$$\mathcal{E}(mn) \rightarrow \mathcal{E}(m) \vee \mathcal{E}(n)$$

- We use $\mathcal{E}$ to denote "x is even"

Note: the domain is integers

- is difficult to prove.

The contrapositive of 1. is

$$\neg(\mathcal{E}(m) \vee \mathcal{E}(n)) \rightarrow \neg(\mathcal{E}(mn))$$

this simplifies to:

$$\mathcal{O}(m) \wedge \mathcal{O}(n) \rightarrow \mathcal{O}(mn)$$

more hypotheses that 1. and a simpler conclusion

We now prove 1. directly (Less verbosety)

1. $\mathcal{O}(n)$
2. $\exists k(n = 2k + 1)$
3. $k = 2l + 1$
4. $\mathcal{O}(m)$
5. $\exists k(m = 2k + 1)$
6. $m = 2o + 1$
7. $mn = (2l + 1)(2o + 1) = 2(2lo + l + o) + 1$

Since $\mathcal{O}(m) \wedge \mathcal{O}(n) \rightarrow \mathcal{O}(mn)$

We know by contraposition, that $\mathcal{E}(mn) \rightarrow \mathcal{E}(m) \vee \mathcal{E}(n)$

Proof by contradiction

To prove p , we can instead show that $\neg p \rightarrow F$

Definition A real number r , is rational so that $r = \frac{p}{q}$, (Note: $q \neq 0$)

Claim

$$\exists r (r \text{ is rational})$$

We'll prove the claim by giving an example,

Lemma

The square root of 2 is not rational.

Literally, the claim is saying two appropriate numbers, p and q , do not exist.

The negation of the **Lemma** is much more concrete.

Proof of Lemma

Assume (for contradiction (now be careful, we hope something goes wrong, we will blame it on the assumption.)) that $\sqrt{2}$ is rational

Thus $\sqrt{2} = \frac{p}{q}$ for some integers, p and q not both even.

Thus $\sqrt{2} = \frac{p}{q}$ so $2 = \frac{p^2}{q^2}$

and

$$2q^2 = p^2$$

$2q^2$ this is even

p^2 this is a product

Thus by our previous example, (p is even) or (p is even)

$\therefore p$ is even

so $p = 2l$ for some l

and thus

$$2q^2 = (2l)^2 \text{ or}$$

$$2q^2 = 4l^2 \text{ or}$$

$$q^2 = 2l^2 \text{ (this is even)}$$

(q is a product)

Thus (q is even) or (q is even)

so q is even.

This contradicts p and q are not both even (our something went wrong)

$\therefore$ the assumption is false

and

$\neg(\sqrt{2} \text{ is rational})$ is true.

Fact

Any rational number can be written in an equivalent form so it's numerator and denominator are not both even